Advanced Computer Systems Engineering Laboratory (ENCS5150)

EXP 4 – TODO 2 – EditCustomerActivity and Customer Deletion

Overview

In this todo, you will **extend your Experiment 4 project** to Create the **EditCustomerActivity** for **editing an existing customer.**
Additionally, you will add options to**: Delete specific customer, delete all customers.**

Task Description

1. **Editing an Existing Customer**
    - Create a new activity called **EditCustomerActivity,** this activity will be used only for **editing an existing customer.**
    - In **MainActivity**, display all customers inside **containers** (e.g., LinearLayouts, or clickable boxes). When the user **clicks on any customer container or edit button inside the container**, open **EditCustomerActivity**, but this time **pass the customer ID** through an **Intent extra**.
    - Inside **EditCustomerActivity**, check whether an ID was received:
        - If yes, fetch the customer details from the database using that ID.
        - Fill all input fields with the existing data.
        - The **ID field** must be **disabled** (not editable).
    - At the bottom of the screen, display a **button labeled "Edit"**.
    - When the user clicks **Edit**, update the customer information in the database and return to **MainActivity**.
2. **Deleting a Specific Customer**
    - In the layout used to display each customer, add a **"Delete" button**, When the user clicks this button, the app should **delete only that specific customer** from the database.
    - After deletion, refresh the customer list immediately so that the change appears on the screen then Show a toast message (e.g., "Customer deleted successfully").
3. **Deleting All Customers**
    - In the MainActivity, add another button labeled **"Delete All Customers"**, this button should be clearly visible (for example, at the top or bottom of the main screen).
    - When the user clicks it, the app should **remove all customers** from the database at once then Show a toast message (e.g., "All customers deleted").

Deadline: After 30 minutes from the start of the ToDo.

Notes

- You must **handle all possible errors and edge cases** (use toast to inform), such as
  - Attempting to add a customer with a duplicate ID.
  - Leaving required fields empty.
  - Database operation failures.
- Use **Pixel 3a XL device (Graphic = Software)**.
- This TODO is **individual work**. Cheating or using LLM Models will result in **0 marks**.

Marks Distribution (Total: 10 Marks)

Component	Description	Marks
Functionality	Implementation of all required features; buttons, game logic, navigation, and correct handling of user actions	4
Error-Handling & Stability	Proper handling of invalid inputs, prevention of crashes, smooth app behavior under different scenarios	2
Design & User Interface	Clear, visually appealing layout, adherence to color schemes, hints, and layout requirements	2
Usability & User Experience	Ease of use, intuitive navigation, responsiveness, and consistency between activities	1
Documentation & Naming Conventions	Correct app name format (`ID_FirstName_LastName`), proper project structure, meaningful variable names, comments …	1

Submission Requirements

Submission through Ritaj

1. **Project.zip file** (Size in KB)
   - From Android Studio: File → Export → Export to Zip File
2. **app-debug.apk file** (Size in MB)
   - From Android Studio: Build → Build Bundle(s) / APK(s) → Build APK(s)
   - You will find the APK file under: app\build\outputs\apk\debug\app-debug.apk

   Send both the APK file and the ZIP file as a reply to my message.

Deadline: After 30 minutes from the start of the ToDo.